

SHOP CUSTOMER DATA

Kaggle Data Analysis Documentation

Python • Pandas • Matplotlib • Seaborn

1. Project Overview

This project analyzes customer information from a shop customer dataset using Python in a Kaggle Notebook. The purpose is to understand the structure of the customer data, calculate basic customer statistics, explore gender and profession distributions, examine age distribution, visualize the relationship between annual income and spending score, study numerical correlations, and identify the top customers by spending score.

2. Project Objective

Load the customer dataset into a Pandas DataFrame.

Inspect the rows, columns, data types, and missing values.

Calculate average age, average annual income, and average spending score.

Analyze customers by gender and profession.

Visualize the distribution of customer ages.

Study annual income versus spending score.

Create a correlation heatmap for numerical variables.

Display the top 10 customers according to spending score.

3. Dataset Description

The dataset contains 2,000 customer records and 8 columns. It provides demographic, income, spending, work-experience, and family-size information.

Column	Meaning	Data Type	Missing Values
CustomerID	Unique customer identifier	int64	0
Gender	Customer gender	object	0
Age	Customer age	int64	0
Annual Income (\$)	Annual income in dollars	int64	0
Spending Score (1-100)	Customer spending score	int64	0
Profession	Customer profession	object	35
Work Experience	Work experience in years	int64	0
Family Size	Number of family members	int64	0

4. Dataset Summary

Metric	Value
Total customers	2,000
Total columns	8
Average age	48.96 years
Average annual income	\$110,731.82
Average spending score	50.96
Average work experience	4.10 years
Average family size	3.77
Missing Profession values	35

5. Tools and Libraries

Python — programming language used for analysis.

Pandas — used for loading, inspecting, sorting, and analyzing tabular data.

Matplotlib — used for creating charts and histograms.

Seaborn — used for statistical and categorical visualizations.

6. Kaggle Notebook Workflow

Import libraries: Pandas, Matplotlib, and Seaborn are imported.

Load dataset: Customers.csv is loaded with `pd.read_csv()`.

Inspect data: The notebook checks the first rows, shape, column names, information, missing values, and descriptive statistics.

Calculate averages: Mean values are calculated for age, annual income, and spending score.

Gender analysis: Gender counts and a gender count plot are produced.

Profession analysis: Profession counts and a profession count plot are produced.

Age analysis: A histogram is used to show the age distribution.

Income vs spending: A scatter plot compares annual income with spending score, separated by gender.

Correlation analysis: A heatmap displays correlations among numerical columns.

Top customers: The 10 customers with the highest spending scores are displayed.

7. Visualizations Included

Customers by Gender — count plot.

Customers by Profession — horizontal count plot.

Age Distribution — histogram.

Annual Income vs Spending Score — scatter plot with gender.

Correlation Heatmap — numerical-variable correlation matrix.

8. Important Data Quality Note

The dataset contains 35 missing values in the Profession column. The current notebook checks and reports missing values, but it does not yet fill or remove those missing Profession entries. This should be considered if the dataset is later used for machine-learning modeling.

9. Key Findings from the Dataset

The dataset has 2,000 customers. The average customer age is 48.96 years, the average annual income is \$110,731.82, and the average spending score is 50.96. The dataset includes 2 gender categories and 9 distinct profession values (excluding missing values).

10. Conclusion

This Kaggle project provides a basic exploratory analysis of shop customer data. The notebook successfully covers data loading, inspection, descriptive statistics, customer segmentation views, and several useful visualizations. The analysis can be extended in the future with data cleaning, customer segmentation, machine-learning models, and more detailed business insights.

11. Possible Future Improvements

Handle the missing Profession values using an appropriate data-cleaning strategy.

Create customer segments based on income and spending score.

Compare spending behavior across professions and age ranges.

Analyze work experience and family size against spending behavior.

Build a machine-learning model for customer segmentation or prediction.

12. Dataset Loading Code

```
df = pd.read_csv("/kaggle/input/datasets/mudasir123312/shop-customer-data/Customers.csv")
```